

Relational Separation Logic for Compiler Verification

Gabriel Desfrene

Supervised by Xavier Leroy and François Pottier

September 2026

Compilers Can Introduce Bugs

Compilers Can Introduce Bugs



Compilers Can Introduce Bugs



Compilers Can Introduce Bugs



Compilers Can Introduce Bugs



CakeML
ML $\rightarrow$ machine

CompCert
C $\rightarrow$ machine

Jasmin
crypto $\rightarrow$ machine

Behavioral Refinement

$$S \models \textit{Spec} \quad \Longrightarrow \quad T \models \textit{Spec}$$

Behavioral Refinement

$$S \models \textit{Spec} \implies T \models \textit{Spec}$$

$$b \in \{\text{Term}(v, m) \mid \text{Div} \mid \frac{1}{4}\}$$

Behavioral Refinement

$$S \models \textit{Spec} \quad \Longrightarrow \quad T \models \textit{Spec}$$

$$b \in S \quad \Longleftrightarrow \quad b \in T$$

$$b \in \{\text{Term}(v, m) \mid \text{Div} \mid \zeta\}$$

Behavioral Refinement

$$S \models \text{Spec} \quad \Longrightarrow \quad T \models \text{Spec}$$

$$b \in S \quad \Longleftrightarrow \quad b \in T$$

$$\textcolor{brown}{b} \in \textcolor{brown}{S} \quad \Longleftarrow \quad \textcolor{blue}{b} \in \textcolor{blue}{T}$$

$$b \in \{\text{Term}(v, m) \mid \text{Div} \mid \textcolor{brown}{?}\}$$

Behavioral Refinement

$$S \models Spec \quad \implies \quad T \models Spec$$

$$b \in S \quad \iff \quad b \in T$$

$$b \in S \quad \longleftarrow \quad b \in T$$

$$T \leq S \triangleq$$

$$b \in \{\text{Term}(v, m) \mid \text{Div} \mid \zeta\}$$

Behavioral Refinement

$$S \models \text{Spec} \quad \Longrightarrow \quad T \models \text{Spec}$$

$$b \in S \quad \Longleftrightarrow \quad b \in T$$

$$b \in S \quad \Longleftarrow \quad b \in T$$

$$T \leq S \triangleq$$

$$b \in S \vee \textcolor{blue}{\not\in} S \quad \Longleftarrow \quad b \in T$$

$$b \in \{\text{Term}(v, m) \mid \text{Div} \mid \textcolor{blue}{\not\in}\}$$

Simulation Relations

$$t_0 \rightarrow t_1 \rightarrow \cdots \rightarrow \text{Term}(v, m)$$

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow \text{Term}(v, m)$$

Simulation Relations



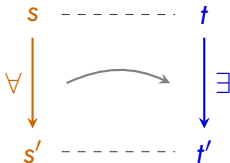
Simulation Relations

Forward



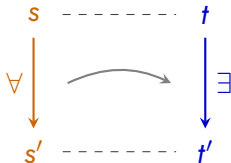
Simulation Relations

Forward



Simulation Relations

Forward



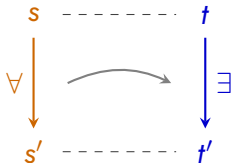
+ Determinism



$$T \leq S$$

Simulation Relations

Forward



+ Determinism



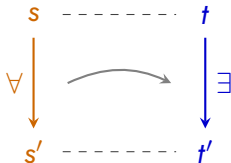
$$T \leq S$$

Backward



Simulation Relations

Forward

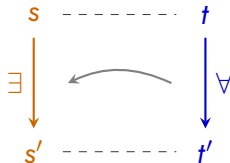


+ Determinism



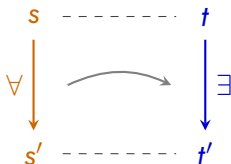
$$T \leq S$$

Backward



Simulation Relations

Forward

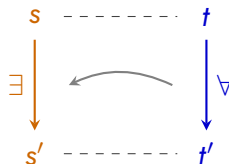


+ Determinism



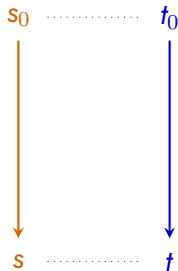
$$T \leq S$$

Backward

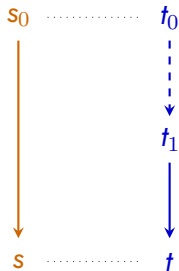


$$T \leq S$$

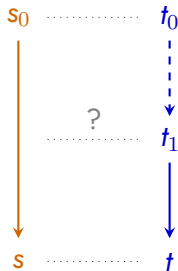
The Stuttering Problem



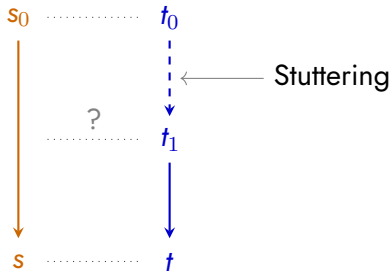
The Stuttering Problem



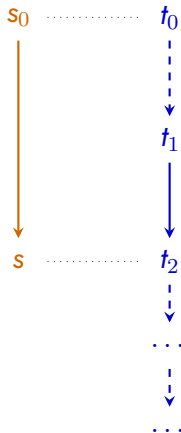
The Stuttering Problem



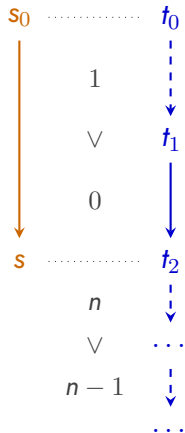
The Stuttering Problem



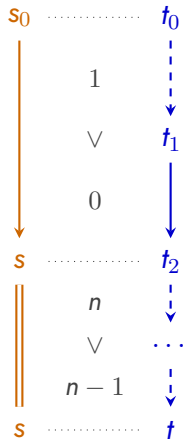
The Stuttering Problem



The Stuttering Problem



The Stuttering Problem



Limitations of Existing Approaches

Limitations of Existing Approaches

Limitations of Existing Approaches

Non-Determinism

Limitations of Existing Approaches

Non-Determinism

CakeML

Not Supported

Limitations of Existing Approaches

Non-Determinism

CakeML

Not Supported

CompCert

External only

Limitations of Existing Approaches

Non-Determinism

CakeML

Not Supported

**Backward
Simulation**

CompCert

External only

Limitations of Existing Approaches

	Non-Determinism	Memory Reasoning
CakeML	Not Supported	
CompCert	Backward Simulation	

Limitations of Existing Approaches

	Non-Determinism	Memory Reasoning
CakeML	Not Supported	Global Invariants
CompCert	External only	Global Invariants

**Backward
Simulation**

Limitations of Existing Approaches

	Non-Determinism	Memory Reasoning
CakeML	Not Supported	Global Invariants
CompCert	External only	Global Invariants

Backward Simulation

Separation Logic

Our Contributions

Our Contributions

Free Backward Simulation

Our Contributions

Relational Separation Logic

Free Backward Simulation

Our Contributions

*Language
Dependent*

*Language
Independent*

Relational Separation Logic

Free Backward Simulation

Our Contributions

*Language
Dependent*

Low level language: μ RTL

*Language
Independent*

Relational Separation Logic

Free Backward Simulation

Our Contributions

Program Transformations

Low level language: μ RTL

Language
Dependent

Language
Independent

Relational Separation Logic

Free Backward Simulation

Our Contributions

Program Transformations

Tunneling

Low level language: μ RTL

Language
Dependent

Language
Independent

Relational Separation Logic

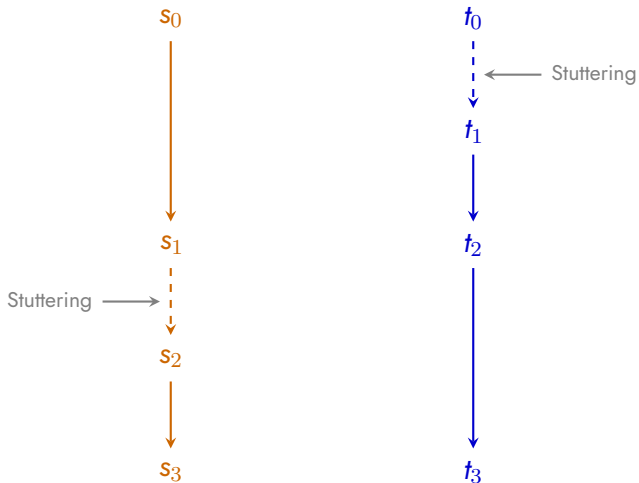
Free Backward Simulation

The Free Simulation

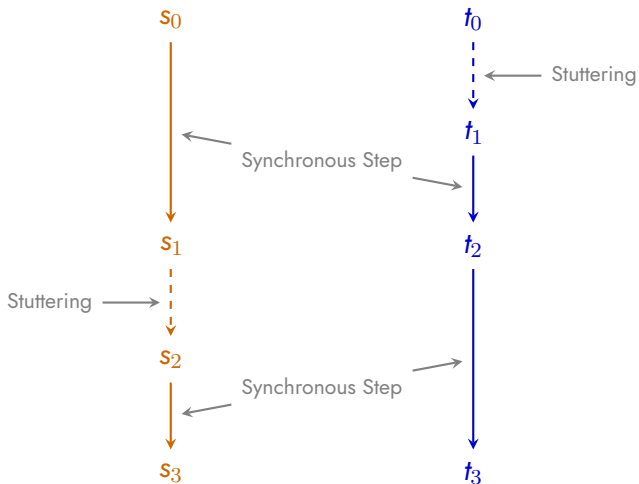
The Free Simulation

 s_0  s_1  s_2  s_3 t_0  t_1  t_2  t_3

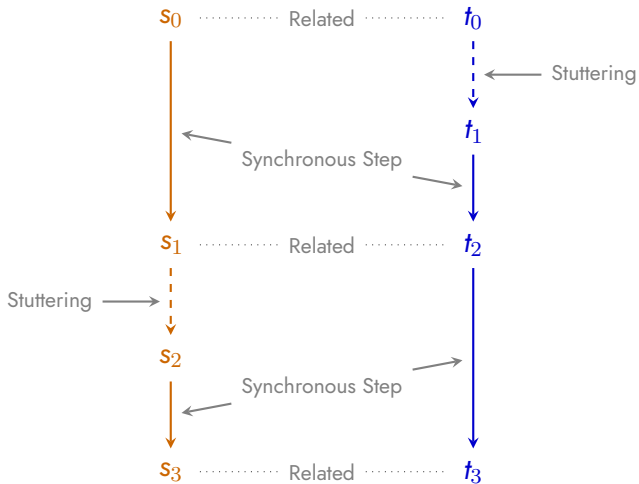
The Free Simulation



The Free Simulation



The Free Simulation



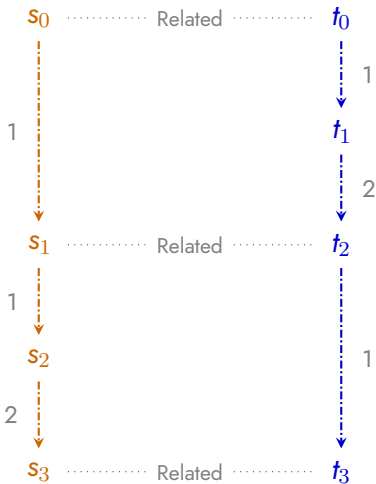
The Free Simulation

 s_0  s_1  s_2  s_3 t_0  t_1  t_2  t_3

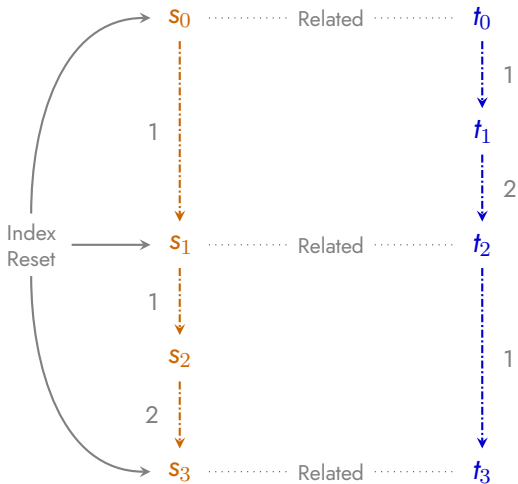
The Free Simulation

 s_0  s_1  s_2  s_3 t_0  t_1  t_2  t_3

The Free Simulation



The Free Simulation



Theoretic Guarantees

Theoretic Guarantees

Sound:

$$\forall ij, \quad \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle \implies T \leq S$$

Theoretic Guarantees

Sound:

$$\forall ij, \quad \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle \implies T \leq S$$

Equivalent:

$$(\exists ij, \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle) \iff T \sim_{\text{explicit}} S$$

Theoretic Guarantees

Sound:

$$\forall ij, \quad \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle \implies T \leq S$$

Equivalent:

$$(\exists ij, \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle) \iff T \sim_{\text{explicit}} S$$

Index Irrelevant:

$$(\exists ij, \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle) \implies \forall ij, \quad \langle T, i \rangle \sim_{\text{free}} \langle S, j \rangle$$

Asymmetric Reasoning

Asymmetric Reasoning

Target

Source

Asymmetric Reasoning

Target

Source



nondeterminism

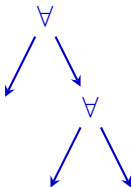
Asymmetric Reasoning

Target

Source



nondeterminism

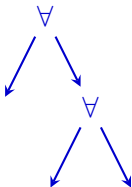


Asymmetric Reasoning

Target



nondeterminism



Source



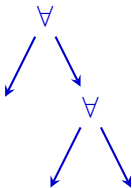
nondeterminism

Asymmetric Reasoning

Target



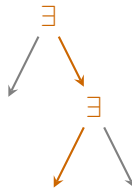
nondeterminism



Source



nondeterminism

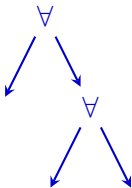


Asymmetric Reasoning

Target



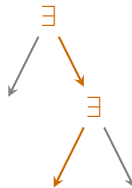
nondeterminism



Source



nondeterminism



$$\text{⚡} \in S \implies T \sim_{\text{free}} S$$

Relational Separation Logic

Values equivalence

Memory Injection

Relational Hoare Quadruples

μ RTL — A Low-Level Intermediate Language

Operational Semantics

Case Study: Eliminating a Redundant Zero-Check

Case Study: Hoisting an Invariant Load

?

Case Study: Short-Circuiting Jump Chains

What Comes Next

Thank you

Questions?